
PEP LLUIS NEGRE

Spain (open to relocation or extended on-site periods for the right role) • pl.negre@outlook.com

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Principal Robotics, AI & Computer Vision Engineer

Experience

Robotics software engineer with 14+ years of experience delivering production-grade autonomous robotic systems operating in harsh, real-world environments. Drove and delivered two world-leading robotics technologies: SubSLAM, a live 3D mapping system for subsea infrastructure, and the world-first autonomous offshore jacket inspection system. Extensive background in hardware-software integration, robot bring-up, data collection pipelines, evaluation tooling, and cross-functional collaboration with mechanical, electrical, and operations teams. Named inventor on multiple robotics and perception patents.

2025 - Current

Principal Robotics, AI & Computer Vision Engineer

Rosenxt UK

Principal Lead for the development of a close-range autonomous inspection solution, driving its architecture and execution end-to-end. Designed hybrid autonomy by combining programmatic behaviours with AI-driven decision-making for robust real-world operation. Developed visual language action models and topological maps to enable intelligent navigation and task execution in complex environments. Delivered a generic, scalable inspection framework adaptable across multiple inspection tasks and diverse asset types.

2024 - 2025

High-Level Autonomy Team Lead

ROVCO Subsea, United Kingdom

Leading the development of world-class AI-driven subsea autonomy. Technically led and delivered the world-first [autonomous inspection of an offshore jacket](#) structure. Driving the end-to-end architecture and algorithmic design — from perception and sensor fusion to decision-making and mission autonomy. Leading a multidisciplinary tech team, setting technical direction, and ensuring delivery of high-impact, production-grade solutions.

2020 - 2024

Principal Computer Vision Engineer

ROVCO Subsea, United Kingdom

Researching and prototyping emerging computer vision technologies to guide subsea exploration and inspection. Designing and developing the core of Rovco's cutting-edge live 3D vision for ROVs and AUVs. Working on tackling challenging underwater imagery problems to keep improving camera pose accuracy and dense 3D models.

2017 - 2020

Senior Computer Vision Engineer

ROVCO Subsea, United Kingdom

Designing online 3D underwater mapping algorithms, based on an accurate Visual-SLAM positioning system and semi-dense 3D reconstructions. Developing cutting-edge solutions for reliable and impressive 3D underwater reconstructions to be executed online during the robot mission. Main contributor and inventor of Rovco's [SubSLAM online 3D mapping](#) sensor for subsea infrastructure. Delivering the most advanced subsea stereo camera system up to date.

2012 - 2017

Postgraduate Researcher

Systems, Robotics and Vision Group, University of the Balearic Islands, Spain

Co-responsible for the entire platform (software and hardware) of the autonomous underwater robot "Turbot". In charge of the whole localization pipeline based on various sensors: IMU, DVL, USBL, GPS and 2 stereo pairs. Specialized in stereo localization, loop closing detection and 3D reconstruction of the environment and the tools to visualize it.

Responsible for the design and development of the operating interface of the vehicle based on HTML5, Google maps API and Rosbridge.

2012 - 2015

CEO at Binibook.com

Binibook S.L.

Lead of the platform to write and distribute books online. In charge of the website and apps developed for iOS and Android. Create value, manage the group and keep developing code in HTML5, Javascript, CSS, PHP, NodeJS, Xcode and Java.

2011 - 2012

Software Developer

Sampol Communications S.L.

Developing drivers for IAS/Wonderware, cameras, relies and many other sensors for factories and intelligent houses.

Implementing power monitoring systems to improve the building efficiency with powerful web front-ends and databases.

Responsible for the project of automatic reading of meters of water and electricity for the Palma de Mallorca Harbour's.

2009 - 2011

Control Engineer

Alstom Power S.L.U.

Responsible for the control design and tuning of the wind turbines ECO74 and ECO80.

Specialized in system identification and fault-tolerant control in adverse weather conditions.

Education

2008 - 2010

Master Degree in Automation and Robotics

University of Catalonia UPC, Spain

Advanced control techniques. Mobile robotics. Computer vision.

Final grade: 8.23/10

2004 - 2008

Industrial Engineer in Industrial Electronics

University of the Balearic Islands, Spain

Computer vision. Robotics. Electronics. Control theory.

Final grade: 6.74/10

2002 - 2004

Bachelor Degree Scientific-technological

IES Guillem Colom Casasnovas, Soller, Spain

Skills

Tools & libraries

- C++ (production robotics systems)
- ROS / ROS2
- Linux (robot deployment)
- Docker (robot + tooling environments)
- Hardware-integrated systems
- NVIDIA Jetson / embedded compute
- Python, CUDA, PyTorch, TensorFlow
- OpenCV, PCL, g2o, Ceres

Robotics

14+ years of experience in robotics. Highly focused on autonomous robots for harsh environments, VLM, SLAM, sensor fusion and stereo vision-based localization systems.

Delivered two world-class subsea technologies: SubSLAM live 3D sensor and offshore jacket structure autonomous inspection.

Software

Languages

Senior programmer in C++, Python, CUDA, PyTorch & TensorFlow.

Experience working in mixed C++ / Rust codebases.

Background in web, C#, Java, typescript and Matlab.

English & Spanish

Relevant Publications

P. L. Negre, F. Bonin-Font, G. Oliver. Global Image Signature for Visual Loop-Closure Detection. In Autonomous Robots, Published Online, Springer US, vol. 40, pp. 1403-1417, December 2016.

P. L. Negre, F. Bonin-Font, G. Oliver. Cluster-Based Loop Closing Detection for Underwater SLAM in Feature-Poor Regions. In IEEE International Conference on Robotics and Automation (ICRA), Stockholm (Sweedden), 2016. **BEST STUDENT PAPER AWARD FINALIST.**

P. L. Negre, F. Bonin-Font, M. Massot, G. Oliver. Stereo-Vision Graph-SLAM for Robust Navigation of the AUV SPARUS II. In IFAC Workshop on Navigation, Guidance and Control of Underwater Vehicles (NGCUV), Girona, 2015.

P. L. Negre, F. Bonin-Font, G. Oliver. Stereo Graph-SLAM for Autonomous Underwater Vehicles. In Intelligent Autonomous Systems 13 (Proceedings of the 13th International Conference on Intelligent Autonomous Systems IAS13), Padova/Venice, pp. 351-360, 2014.

Patents

- PL. Negre, E.D. Ruiz. 2023. Computing device and method for mapping. EP4471722A1. Filed May 31, 2023, published Dec 4, 2024 (pending).
- I. Wallace, PL. Negre, L. Hill. 2021. Computing device and method for video object detection. EP3651056. Filed Nov 6, 2018, and issued Mar 28, 2021.
- I. Wallace, L. Hill, PL. Negre, N. Read. 2020. Subsea surveying system. GB2582554. Filed Mar 21, 2019, and issued Sep 30, 2020.
- I. Wallace, L. Hill, PL. Negre N. Read. 2020. Surveying system for surveying in harsh environments. EP3712558. Filed Mar 21, 2019, and issued Sep 23, 2020.
- I. Wallace, L. Hill, B. Allen, PL. Negre. 2019. Subsea camera module and multi camera system. GB2570748. Filed Sep 4, 2018, and issued Aug 7, 2019.

Portfolio

Author/inventor or main contributor of:

- World-first AI-driven [autonomous subsea jacket inspection](#) (main inventor).

- [SubSLAM live 3D mapping](#) for subsea infrastructure (main inventor).

- [Stereo Slam](#). A stereo camera SLAM system, based on my paper of ICRA'16. Nominated for the best student paper award (author).

- [Libhaloc](#). A loop closure detection system, based on my Autonomous Robots journal paper (author).

- Contributions to ROS most popular visual odometers: [viso2](#) and [fovis](#).

- 3D mapping and reconstruction:

* Rema (private repo). A full registration and mapping library for stereo cameras, using PnP for image to image registration, g2o for graph optimization and Ceres Solver for bundle adjustment.

* [Uware](#). Simple tool for image to image registration, based on PnP and ICP.

- 3D visualization

* [Pointcloud web viewer](#). A 3D pointcloud web viewer.

- Camera drivers: https://github.com/srv/pg_spinnaker_camera and https://github.com/srv/prosilica_driver